**School of Information Technology and Engineering**

**Advanced PLC**

**“SCM Intro Change”**

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**Almaty 2025**

## Introduction

The purpose of this practice is to study the principles of Sequential Control Modules (SCM) in Honeywell Experion PKS and their interaction with Control Modules. The work focuses on configuring the PIDA block by switching its operating mode to MANUAL and setting the mode attribute to PROGRAM. During the practice, a C300 controller, CEEC300 environment, IOLINK, and required Control Modules were created and configured. Special attention was given to the logical sequence of operations using flags, transitions, and state changes. This practice helps develop practical skills in sequential logic implementation and controller operation in industrial automation systems.

**The goal of our Practice** is to create Sequential Control Module (SCM) by setting Control Module's PIDA.MODE as MANUAL and by setting PIDA.MODEATTR as PROGRAM.

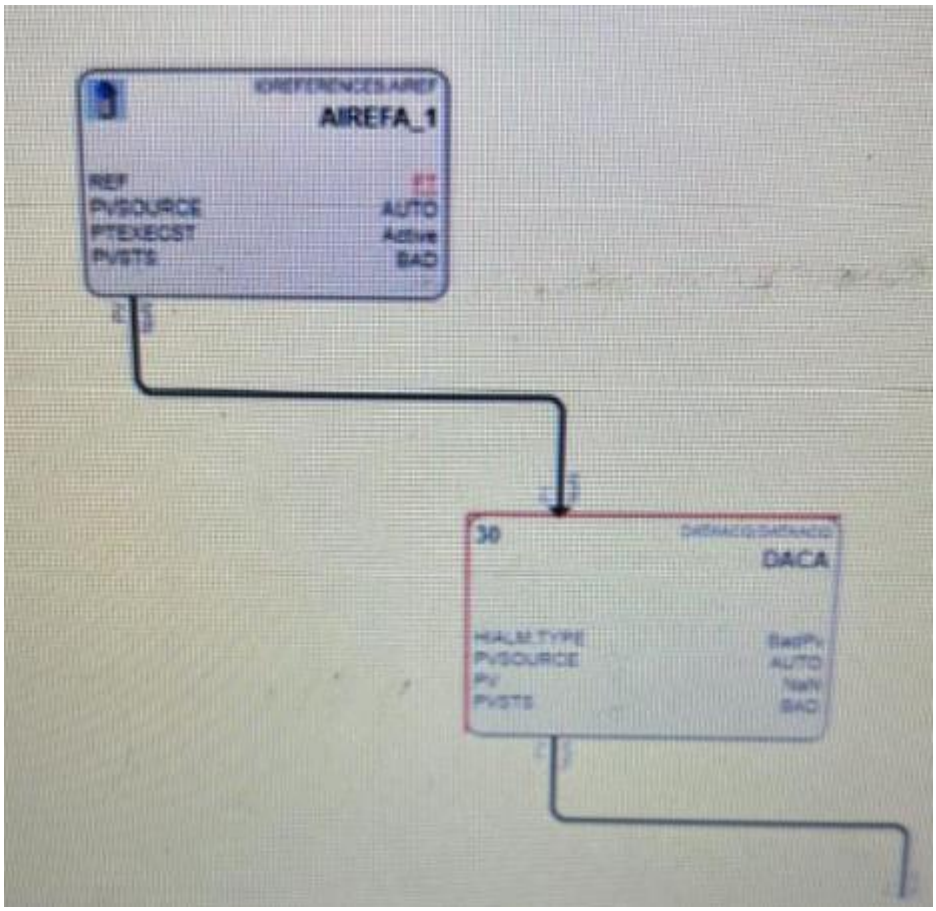
**Steps:**

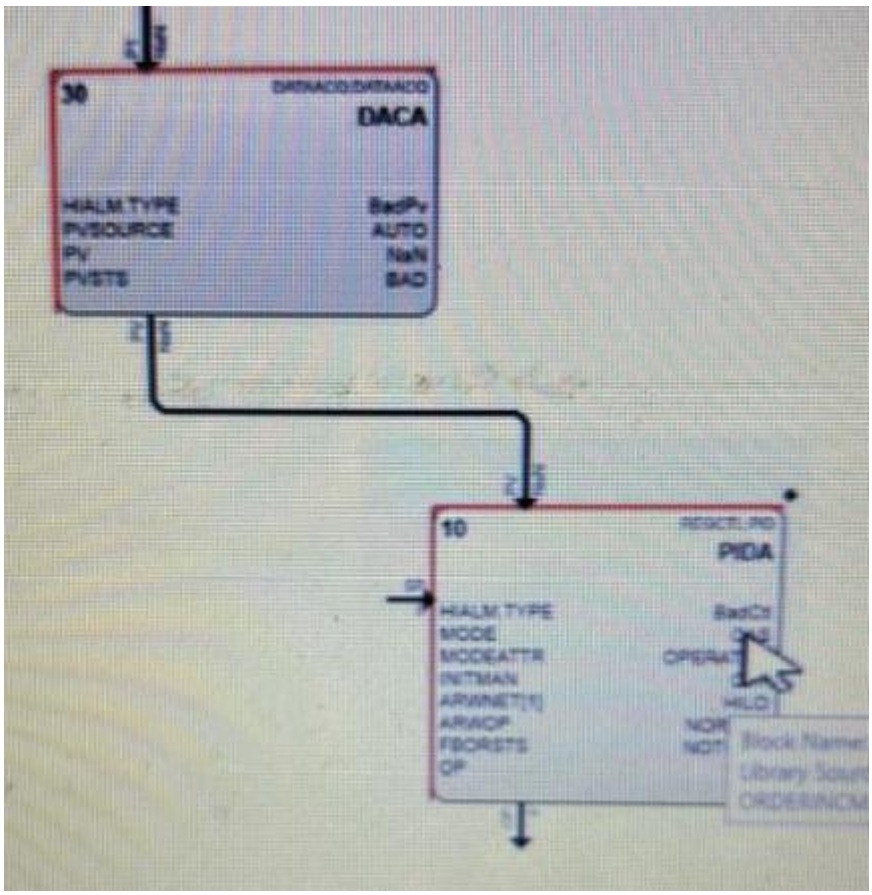
1. First, we created C300 Controller, CEEC300, UIO module for SERIES_C_IO_TYPE IOLINK and Control Modules.
2. Second, we made FC_17, which includes AIREFA_1, DACA and PIDA blocks.
3. Next, we obtained STARTFLAG Control Module with its FLAGA block.
4. Then, we received Sequential Control Module (SCM).
5. We installed Check flag as STARTFLAG.FLAGA.PVFL=1 to InvokeMain Transition Block. Also, we connected Primary and Secondary Gates.
6. We installed FC_17.PIDA.MODEATTR:=2 in STEPA block and set PROGRAM Mode Attribute to it.
7. Thereafter, we installed FC_17.PIDA.MODE:=4 and set MANUAL mode to it.
8. We typed FC_17.PIDA.OP:=0 as next step.
9. We typed FC_17.MODEATTR:=1.
10. So, we renamed Control Module as FCMANUAL.
11. Next, we loaded STARTFLAG and FCMANUAL to Monitoring tab. It worked well.
12. After that, we run FC_17 at Station and got Simulation panel for it.
13. We settled 30% to Output.
14. We turned ON the Actual State Flag of FLAGA Output.

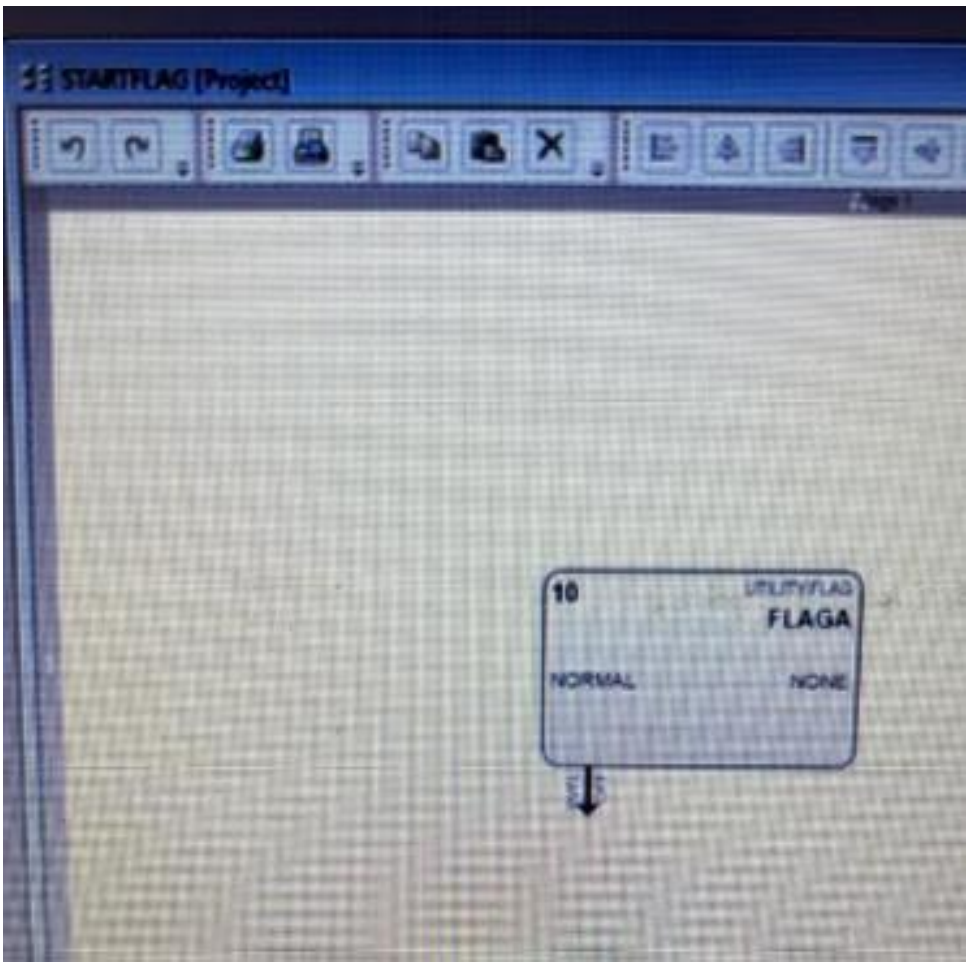
15. We changed states of FCMANUAL by Command (Start, Hold, Active, etc.)

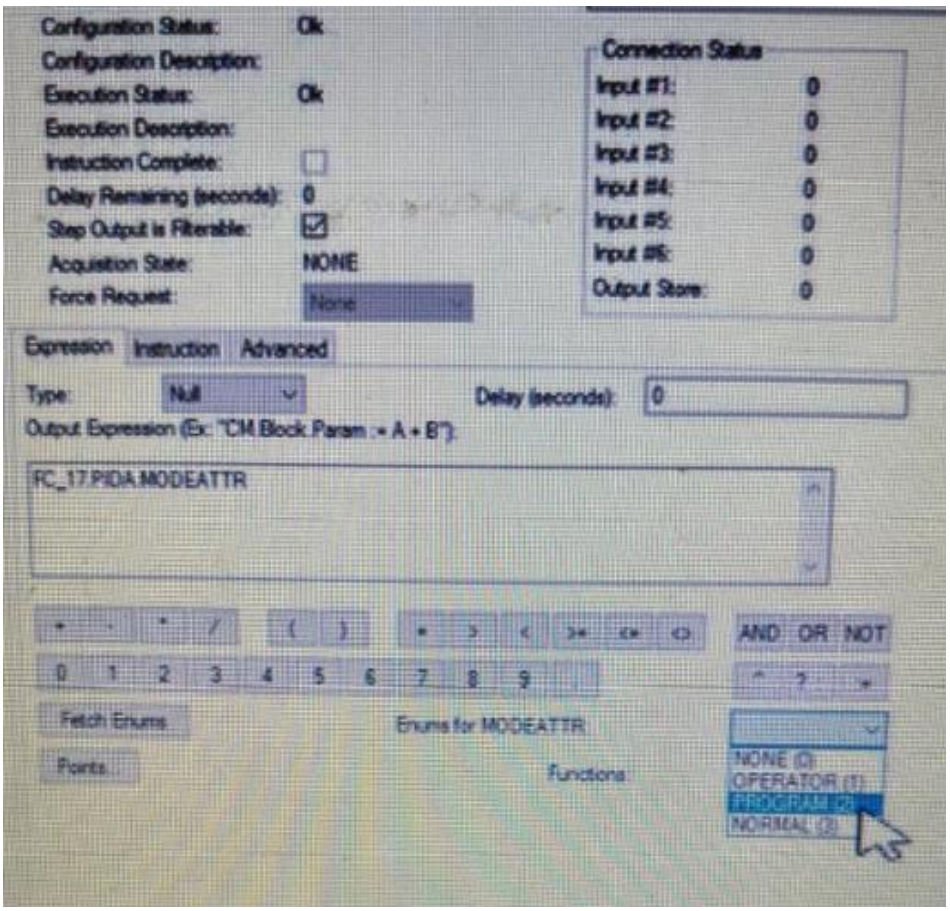
16. Finally, we detected the changing position of FC_17 Output and it was decreasing. We proceeded experimenting by giving Set Points and changing States.

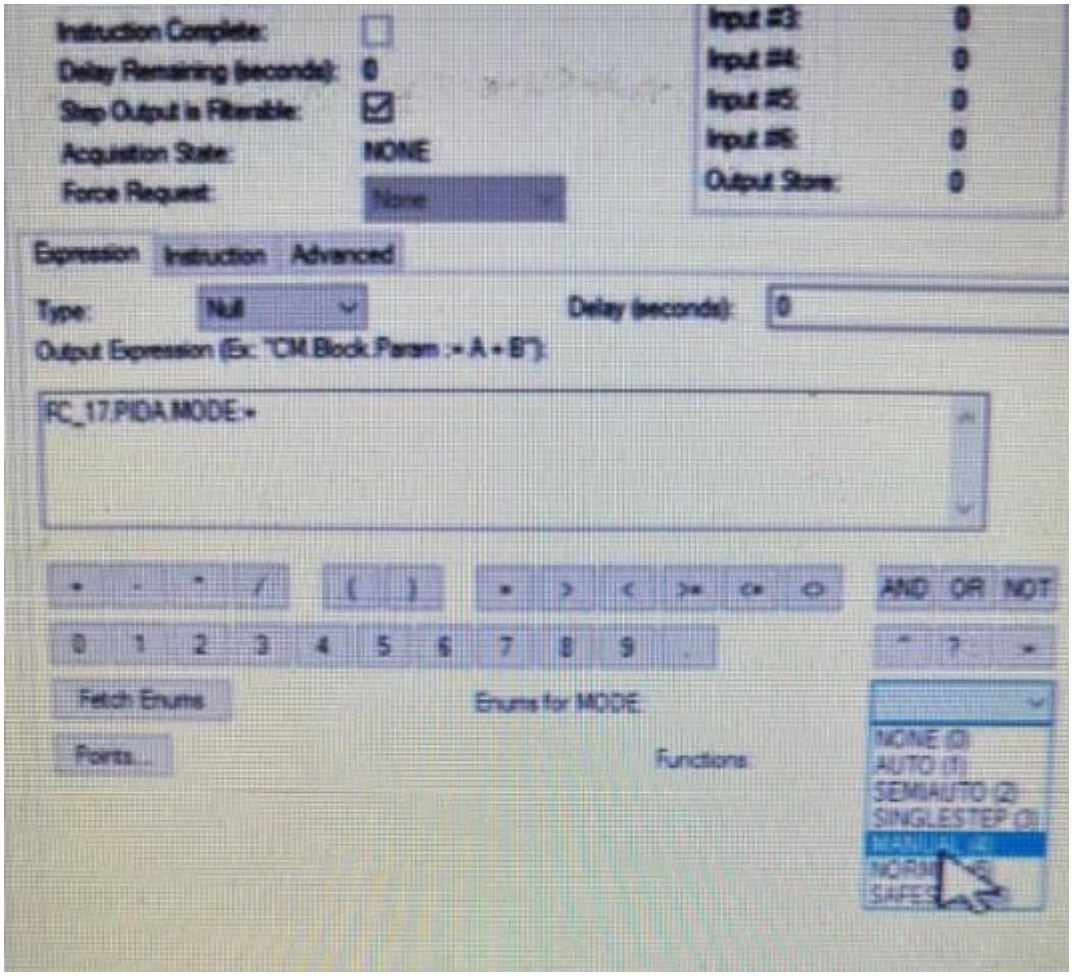
**Mobile photos:**

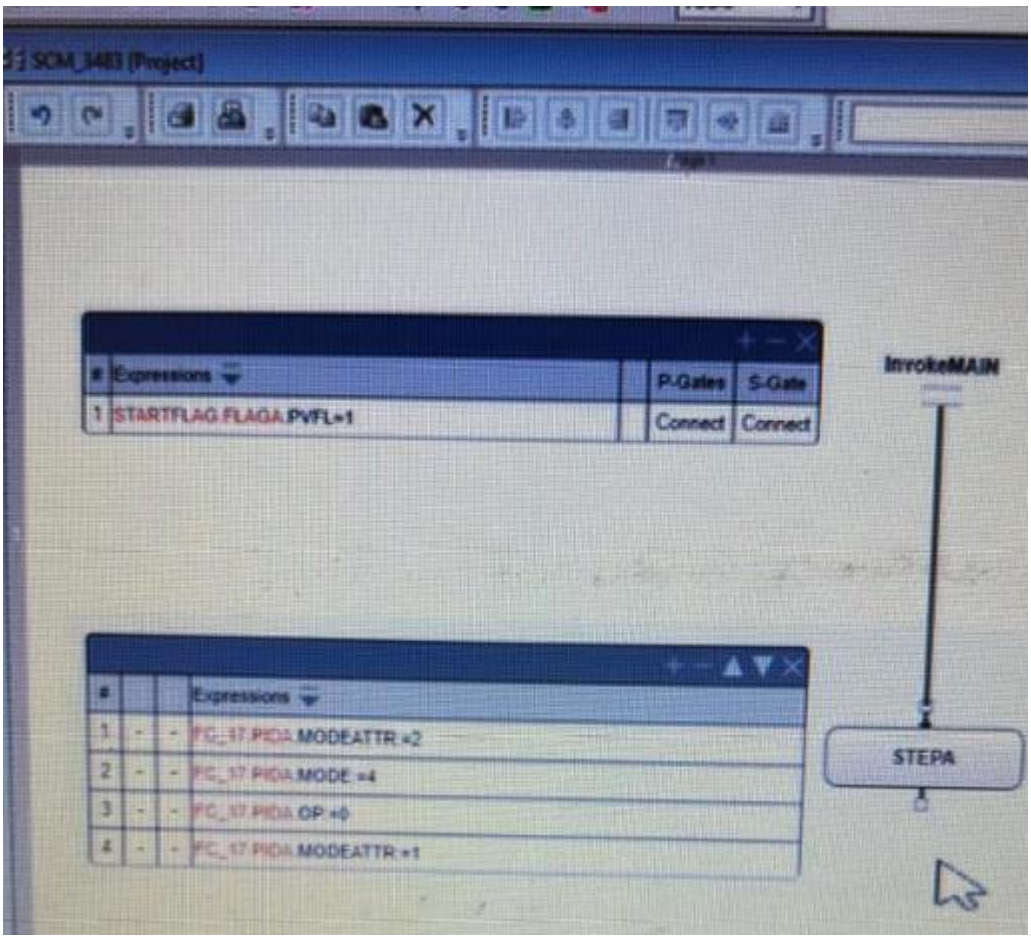


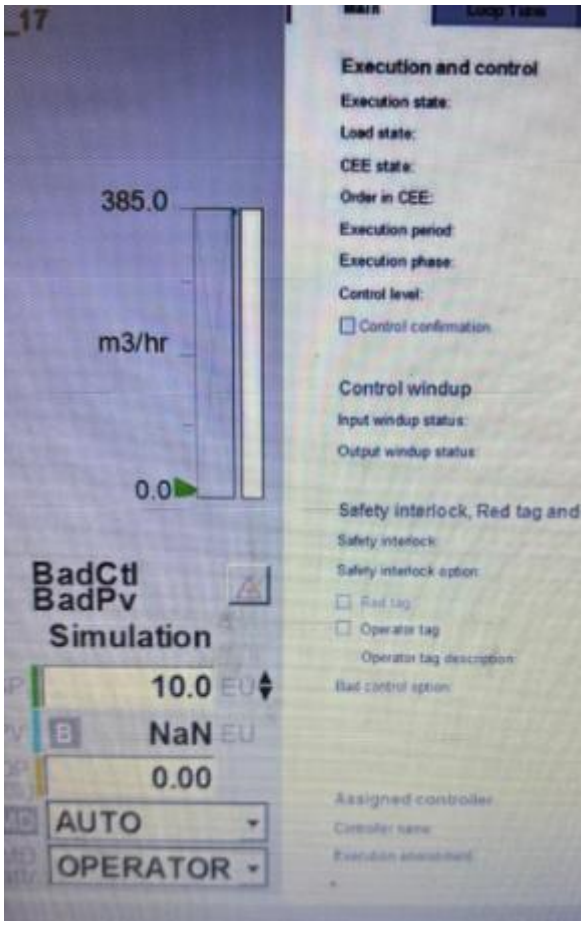

30
DACA
HIALM.TYPE
PVSOURCE
PV
PVSTS
BadPv
AUTO
BAD
10
PIDA
HIALM.TYPE
MODE
MODEATTR
INITMAN
ARWINET[1]
ARWOP
FBORSTS
OP
BadCt
HILO
Block Name

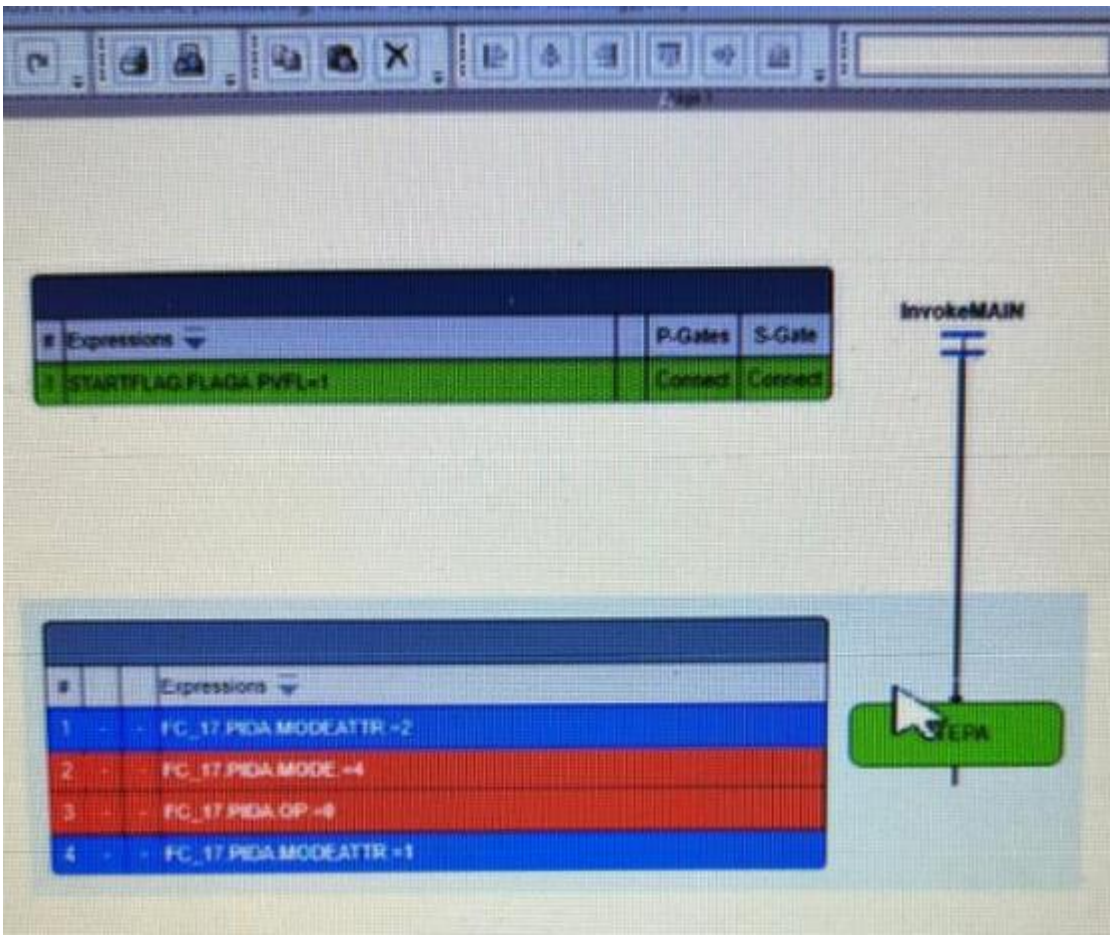

STARTFLAG [Project]
10
UTILITY/FLAG
FLAGA
NORMAL
NONE

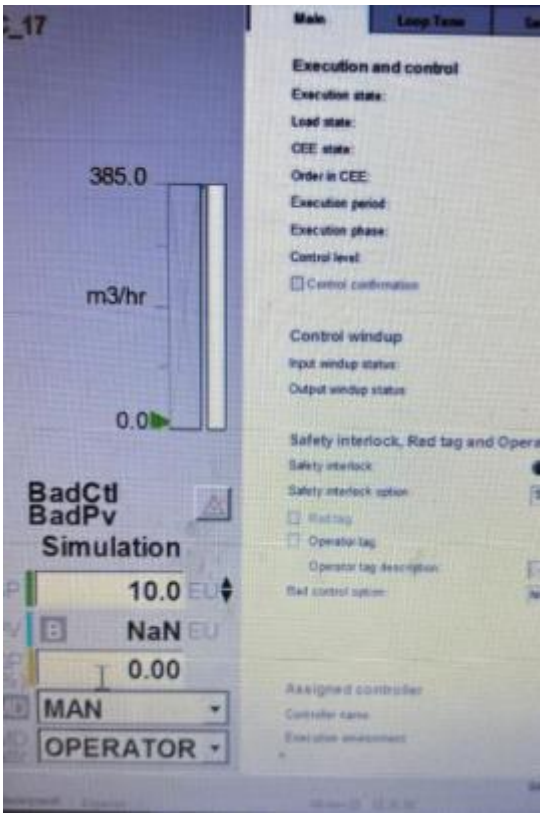

Configuration Status: Ok
Configuration Description:
Execution Status: Ok
Execution Description:
Instruction Complete:
Delay Remaining (seconds): 0
Step Output is Filterable:
Acquisition State: NONE
Force Request: None
Connection Status
Input #1: 0
Input #2: 0
Input #3: 0
Input #4: 0
Input #5: 0
Input #6: 0
Output Store: 0
Expression Instruction Advanced
Type: Null
Delay (seconds): 0
Output Expression (Ex: "CM.Block.Param := A + B"):
RC_17.PIDA.MODEATTR
AND OR NOT
Fetch Enums
Enums for MODEATTR
Points...
Functions
NONE (0)
OPERATOR (1)
PROGRAM (2)
NORMAL (3)


Instruction Complete:
Delay Remaining (seconds): 0
Step Output is Filterable:
Acquisition State: NONE
Force Request: None
Input #3: 0
Input #4: 0
Input #5: 0
Input #6: 0
Output Store: 0
Expression Instruction Advanced
Type: Null
Delay (seconds): 0
Output Expression (Ex: "CM.Block.Param := A + B"):
RC_17.PIDA.MODE :=
AND OR NOT
Fetch Enums
Enums for MODE
Points...
Functions
NONE (0)
AUTO (1)
SEMIAUTO (2)
SINGLESTEP (3)
MANUAL (4)
NORMAL (5)
SAFE (6)


SOM_3483 (Project)
# Expressions
P-Gates
S-Gate
1 STARTFLAG.FLAGA.PVFL=1
Connect
Connect
InvokeMAIN
# Expressions
1 FC_17.PIDA.MODEATTR=2
2 FC_17.PIDA.MODE=4
3 FC_17.PIDA.OP=0
4 FC_17.PIDA.MODEATTR=1
STEPA


_17
385.0
m3/hr
0.0
BadCtl
BadPv
Simulation
10.0 EU
NaN EU
0.00
AUTO
OPERATOR
Execution and control
Execution state:
Load state:
CEE state:
Order in CEE:
Execution period:
Execution phase:
Control level:
Control confirmation
Control windup
Input windup status:
Output windup status:
Safety interlock, Red tag and
Safety interlock:
Safety interlock option:
Red tag
Operator tag
Operator tag description:
Bad control option:
Assigned controller
Controller name:
Execution environment




So, the mobile photos show the process of the Practice.

## Conclusion

As a result of this practice, a fully functional Sequential Control Module was successfully developed and tested. The correct configuration of PIDA.MODE and PIDA.MODEATTR allowed manual control of the process output within the SCM framework. The interaction between Control Modules, flags, and transitions demonstrated the importance of proper sequencing in automation logic. Monitoring and simulation confirmed that output values responded correctly to state changes and operator commands. Overall, this practice enhanced understanding of SCM behavior and reinforced practical experience with Advanced PLC control strategies.